

Yongjae Kwon 권용재

Web Developer · from screens to server and DB

Email	yongjae116@gmail.com	Phone	+82 10-9470-1704
GitHub	github.com/YongjaeKwon	Portfolio	www.yongjaekwon.com
Location	Yongin, South Korea	PDF deck	yongjaekwon.com/portfolio.pdf

I build the features that are needed, and follow them into real use.

I work on live public-sector and B2B business systems: I agree requirements with operations staff and own the work from Vue and WebSquare screens through Spring APIs and SQL to deployment and operations. I start slow queries from the execution plan, make rules hold the same way however many servers run, and prepare a way back before changing production data.

60s → 63-69ms A lookup that never finished within 60 s Education device system · measured on production	300-400 files Attachment zips turned into jobs with visible progress B2B partner portal	108,237 devices Education devices issued QR codes Education device system · of 111,593
---	--	---

Experience

TG Narae LLC · Web Developer · Solutions Team · Full-time

Jun 2024 – Present

On a B2B partner portal and the Seoul Metropolitan Office of Education's device management system, I own requirements, DB modeling, HTTP API and SQL design and development, screens, external integrations, user acceptance, deployment and operations.

Work projects

B2B Partner Portal (PPS)

Feb 2025 – Present

Live · from initial build through features, acceptance, deployment and operations

A portal where headquarters and about 500 partner companies handle partner registration and contracts, plus training, certification, documentation and evaluation for 824 field-engineer (CE) accounts.

- **Zippping 300-400 attachments became a job with visible progress** — Split the zip out of the request into a job-ID based async job. The screen shows waiting / running / done / failed; state lives in a Hazelcast distributed map, so both servers report the same progress and the download resumes after a refresh.
- **Removed the cause of mislinked attachments, blocked on screen and server** — A new CE could inherit the previous CE's attachment group through shared Vue state; each screen now creates fresh state, and the server rejects an attachment group already used by another CE.
- **One password-reset rate limit across two servers** — Atomic acquisition on a Hazelcast distributed map lets only one of any concurrent requests send a code. Limiter keys are SHA-256 hashes, so no raw account or phone number is stored; edge cases are pinned by unit tests.
- **Headquarters account provisioning: 3 steps into one screen** — Internal account, portal account and manual DB entry became one procedure; if the internal-system call fails, the portal save rolls back.
- **Notification sends: 5 send points into one shared service** — Recipients and send conditions are managed in the DB, so a policy change is a configuration update, not a code change.
- **Automated deployment** — Checkout, build, transfer, backup and sequential deployment to two servers run as Jenkins jobs.

Tech Vue · JavaScript · Java 21 · Spring Boot · MyBatis · MariaDB · Oracle · Hazelcast · Jenkins

Education Device Management System (TSMS)

Sep 2025 – Present

Live · maintenance and new program features, acceptance, deployment

Manages about 110,000 education devices for the Seoul Metropolitan Office of Education, from registration through delivery, installation, repairs and inspections. Used by several operations teams and by parents in four education offices.

- **A lookup that never finished within 60 s now returns in 63-69 ms (at least ~870x)** — The execution plan showed the customer filter never reached inside a consolidated view. Rewrote it as base-table joins over only the columns the screen uses, and compared result rows before and after (measured on production, Sep 2026).

- **Bulk registration checks behind QR codes for 108,237 devices** — Serials are checked against production-intake records and prior registrations before saving, and QR codes carry a public ID instead of internal identifiers (of 111,593 devices).
- **School inspections moved on screen — 14,882 checklists at 71 schools** — One flow from schedule approval to checklist, e-signature and confirmation; a version check on save keeps one inspector from overwriting another's results (Sep 2026, of 119 target schools).
- **External API keys removed from the browser on 25 screens** — Map, address and education-administration calls moved to a shared server path, so an integration change is a single server-side edit.
- **Public enrollment screens for parents in 4 education offices** — Consent, delivery booking and QR pickup on a phone without logging in, including deadline locks, error screens and a dark-mode QR fix.
- **Live inspection data cleaned up with a way back** — Prepared backup, rollback and verification SQL first, then removed duplicate rows while keeping results and re-keyed storage to the full serial (applied Jul 2026).

Tech WebSquare5 · JavaScript · JSP · Java 11 · Spring MVC · MyBatis · MariaDB

Personal & team projects

ticket-rush · First-come ticketing system

Aug 2026 – in progress

Personal · public code github.com/YongjaeKwon/ticket-rush

A reservation system built test-first around one rule: the same seat is never sold twice.

- **100 concurrent requests for one seat, 1 success** — A Redis hold reserves the seat for 5 minutes, expired holds are refused at payment, and a MySQL (show, seat) primary key blocks a second confirmation.
- **Still one confirmation when Redis holds disappear** — With hold keys deleted to create 10 duplicate holds confirmed at once, exactly one seat and one booking remain (Testcontainers integration test).
- **Payment retries** — A retry with the same key gets the first response back; a new attempt after a decline uses a new key.
- **Web and E2E** — Next.js seat map (Canvas), queue position (SSE) and payment screens with Playwright E2E in CI. Now building phase 3: relaying outbox events to Kafka.

Tech Java 21 · Spring Boot 4 · Spring Modulith · MySQL 8.4 · Redis 7 · Flyway · Next.js · TypeScript · Testcontainers · Playwright

ONULSAI · Offline dating event service

Sep 2026 – in progress

Personal · private

A service for actually running a Sunday café event where each participant talks one-on-one with seven people and is matched only on mutual choice.

- **20 people apply at once for the last seat, 1 succeeds** — The round row is locked before counting seats by gender; eight concurrent clicks by one member still leave one application, and a 2-second lock timeout keeps the service responsive (PostgreSQL integration tests).
- **Service boundaries** — Member, schedule, matching, notification and payment services sit behind an API gateway, each with its own database; the gateway verifies JWTs and forwards internal headers.
- **Participant and operator web apps** — Sign-up, schedules and applications, plus member lookup, round management and cancellations work today; payment, check-in and matching come next. Each feature starts from an agreed design doc.

Tech TypeScript · NestJS · Next.js · PostgreSQL · Prisma · Turborepo · Vitest · Playwright

ReachRich · Personal investment research platform

Mar 2026 – in progress

Personal · private · public synthetic-data demo github.com/YongjaeKwon/quant-lab

Gathers my brokerage account and market data to validate investment strategies and run them as paper trades. Redesigned in a new repository in Aug 2026, carrying over only validated code, in six areas.

- **Live account lookup and daily candles for the top 200 KRX stocks** — Accounts are replaced per date and candles upserted by stock and date, so reruns never duplicate data.
- **Collection that isolates failures** — One failing stock doesn't stop the rest; above a 30% failure rate the job fails as a whole and alerts on Telegram.
- **React dashboard** — Asset summary, equity curve, holdings and trading journal on separate routes; polling pauses in hidden tabs. 205 Python and 96 web tests run in CI (Aug 2026).

Tech Python · FastAPI · SQLAlchemy · SQLite · Parquet · React · TypeScript · GitHub Actions

SSAFY team projects · Frontend

Jan 2023 – May 2023

SSAFST	A tool to write API specs and run requests and load tests (team of 6). Built repeatable, nested input forms with React Hook Form and the request and result screens.	Apr – May 2023
ddoing	An English-vocabulary service where a drawing, judged by AI, moves learning on. Canvas drawing input, judge-server integration, and a double-running timer fix.	Feb – Apr 2023
MODAC	A study-log and study-group service (team of 6). Study-room entry, invite-code checks and the chat screen with Vue 3 and Pinia.	Jan – Feb 2023

Skills

Frontend	JavaScript · TypeScript · Vue · React · Next.js · WebSquare5 · JSP · React Hook Form · TanStack Query · Pinia
Backend	Java · Spring Boot · Spring MVC · MyBatis · Python · FastAPI · NestJS
Data	Oracle · MariaDB · MySQL · PostgreSQL · SQLite · Redis · Hazelcast
Test · Delivery	Vitest · Testing Library · Playwright · JUnit · Testcontainers · pytest · Git · GitHub Actions · Jenkins · SVN · Linux · Tomcat

Education & certification

Sep 2024	SQLD (SQL Developer) · Korea Data Agency
Jul 2022 – Jun 2023	Samsung Software Academy for Youth (SSAFY), 8th cohort · Web development track
Mar 2018 – Aug 2020	Ajou University · B.A. in e-Business (transfer)
Jan 2014 – May 2015	California State University, Chico · Business Administration (E-Business), before transfer